Module 4

Configuring Network Addressing

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Learning Objectives

- Explain IPv4 addressing schemes.
- Explain IPv4 forwarding.
- Configure IP networks and subnets.
- Use appropriate tools to test a host’s IP configuration.
- Explain IPv6 addressing schemes.
- Troubleshoot IP networks and hosts.

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Lesson 4.1

Internet Protocol Basics

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Think About It

What is the purpose of a package mailing label?

What information is important to the delivery company?

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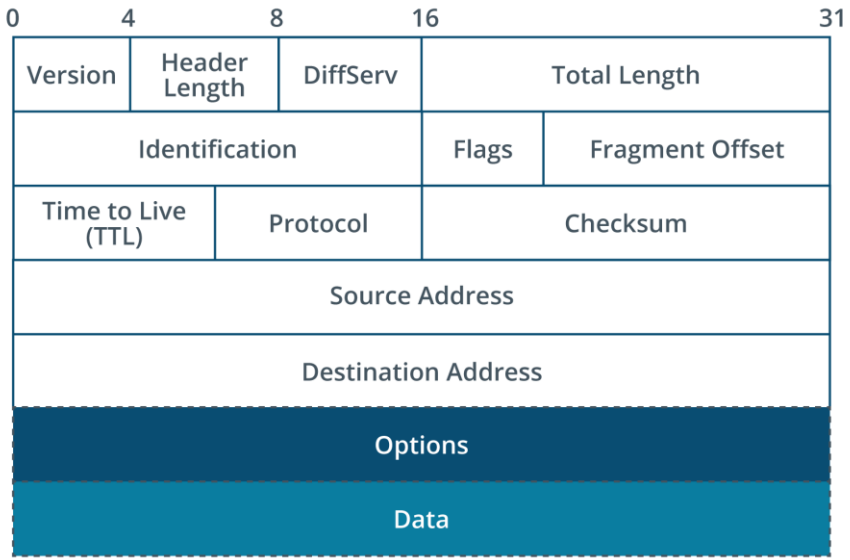
IPv4 Datagram Header



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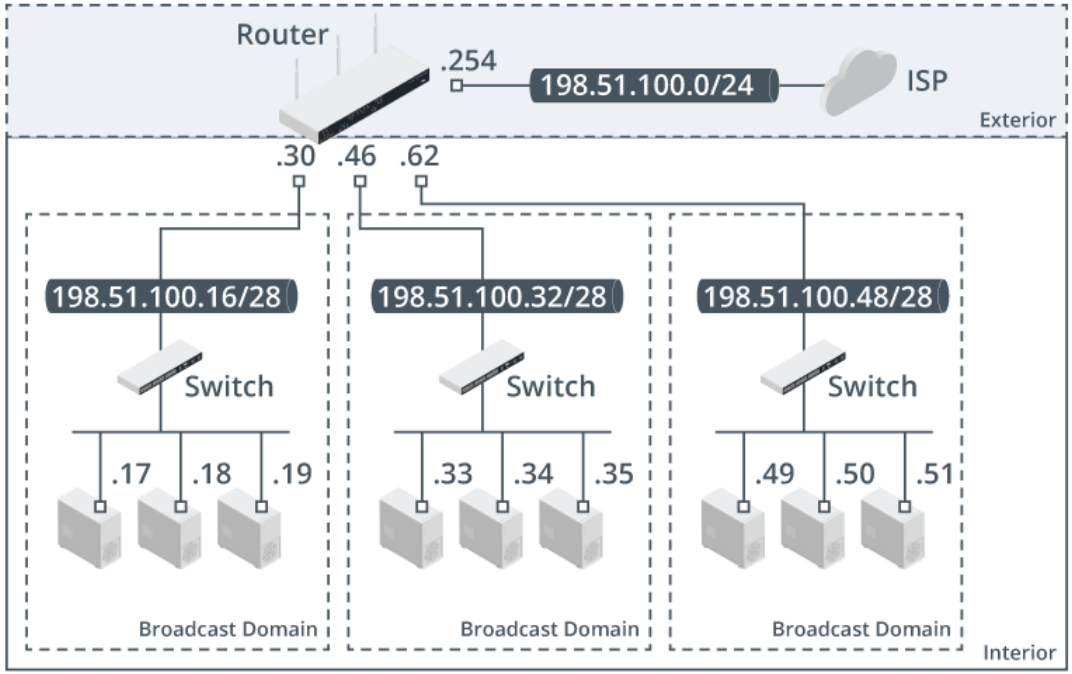
Addressing and Forwarding



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Address Resolution Protocol (ARP)

I have a delivery for
a Juanita Valdez

Let me find
her for you!

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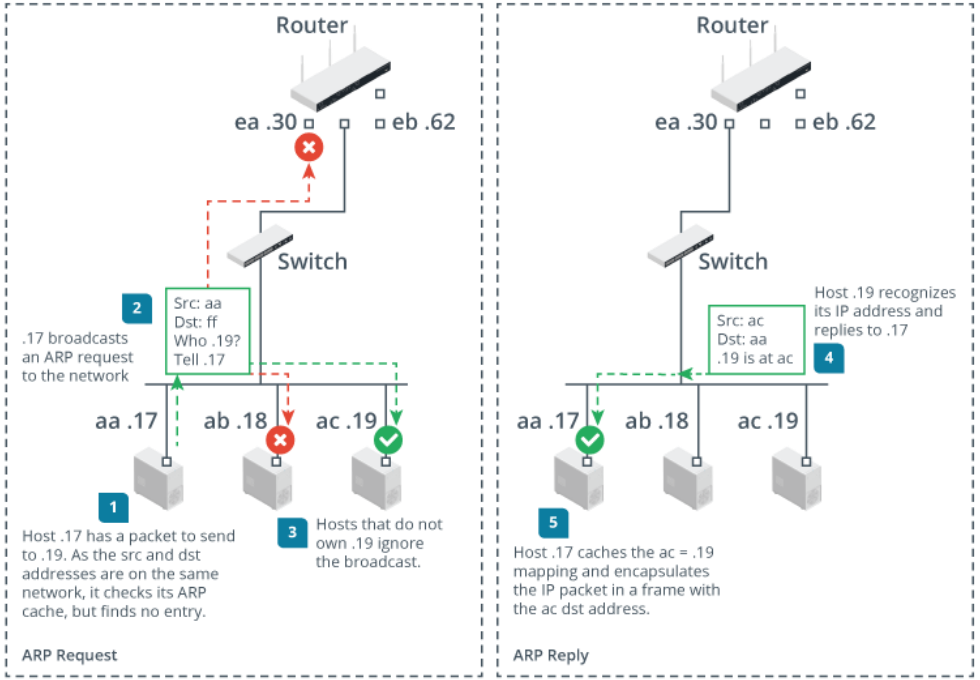
Access Subnet vs. Remote Using ARP



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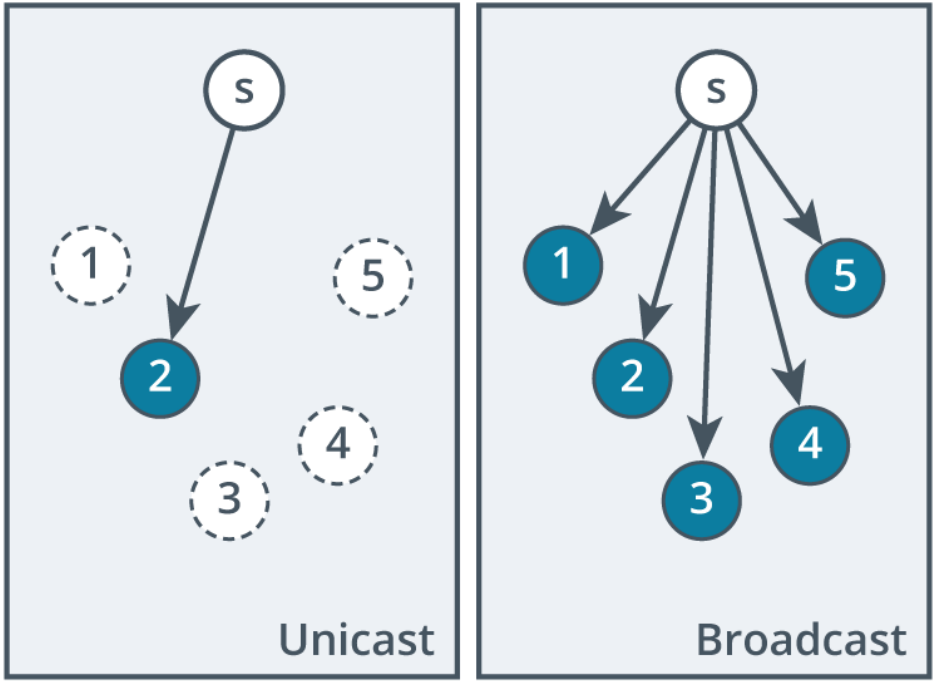
Unicast and Broadcast Addressing



Unicast

- Addressed only to the destination IP
- Sends only to the destination

Broadcast

- Addressed to the last IP address in a network or subnet
- Sends to all hosts on a network or subnet

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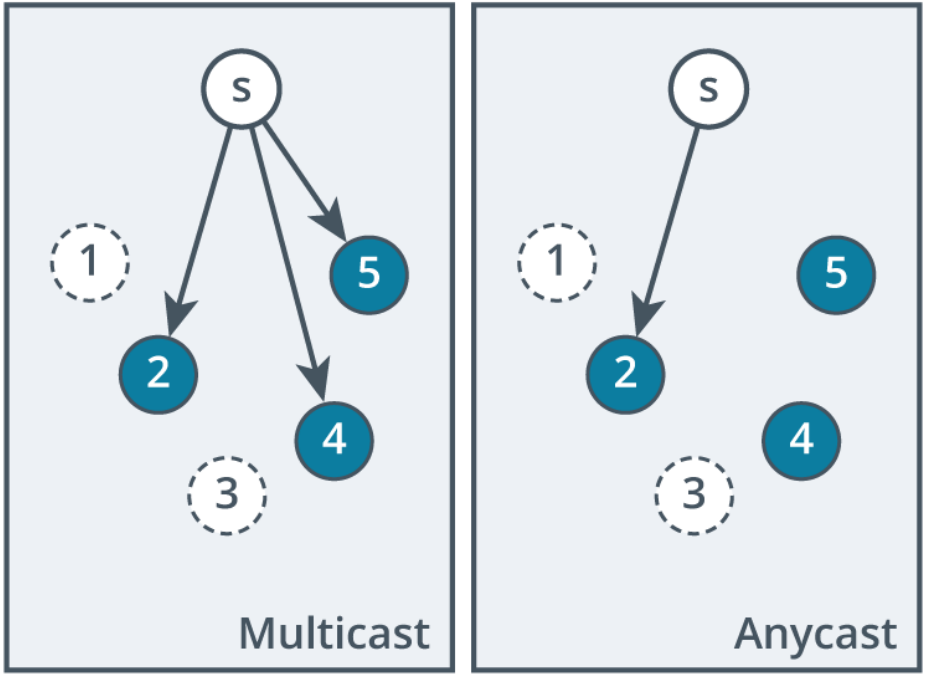
Multicast and Anycast Addressing



Multicast

- Addressed to a specific group of host IP addresses
- Sends to all hosts on a network or subnet

Anycast

- Addressed to any one device out of a group of devices, typically the closest one
- Used for load balancing and redundancy

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Activity: Two Truths and a Lie

IPv4 headers contain a "Destination" field used to route data to the nearest server, regardless of its final destination.

ARP functions like an online phone directory, associating IP addresses with MAC addresses to ensure data reaches the correct device.

Broadcast addressing targets all hosts within a specified local network

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Lesson 4.2

IPv4 Addressing

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IPv4 Addressing

172	.	17	.	154	.	2
101001100		00010001		10011010		00000010
8 bits		8 bits		8 bits		8 bits

IPv4 addresses are decimal representations of four sets of binary octets. Each octet is 8 bits, making all IPv4 addresses 32 bits.

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Activity: Binary Conversion

Convert 11000011 into standard form:

0	1	1	0	0	0	0	1
128	64	32	16	8	4	2	1

Answer: 64 + 32 + 1 = 97

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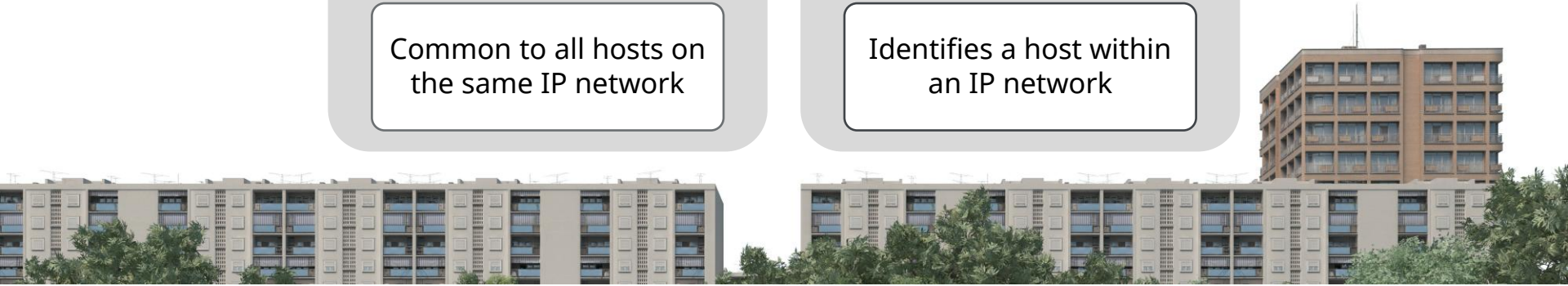
IP Address Formatting

Network ID

- Like an apartment building number
- Common to all hosts on the same IP network

Host ID

- Like an apartment number within a building
- Identifies a host within an IP network



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Subnet Masks

Subnet Masks

- Identifies the network portion of an IP address
- If network ID matches the local network, transmission is sent directly to host – otherwise it is forwarded
- Help devices determine which network or subnet it belongs

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Determining Subnet Mask

Example:

Network: 192.168.1.x
Binary: 11000000.10101000.00000001.00000000

Mask: 255.255.255.0
Binary: 11111111.11111111.11111111.00000000

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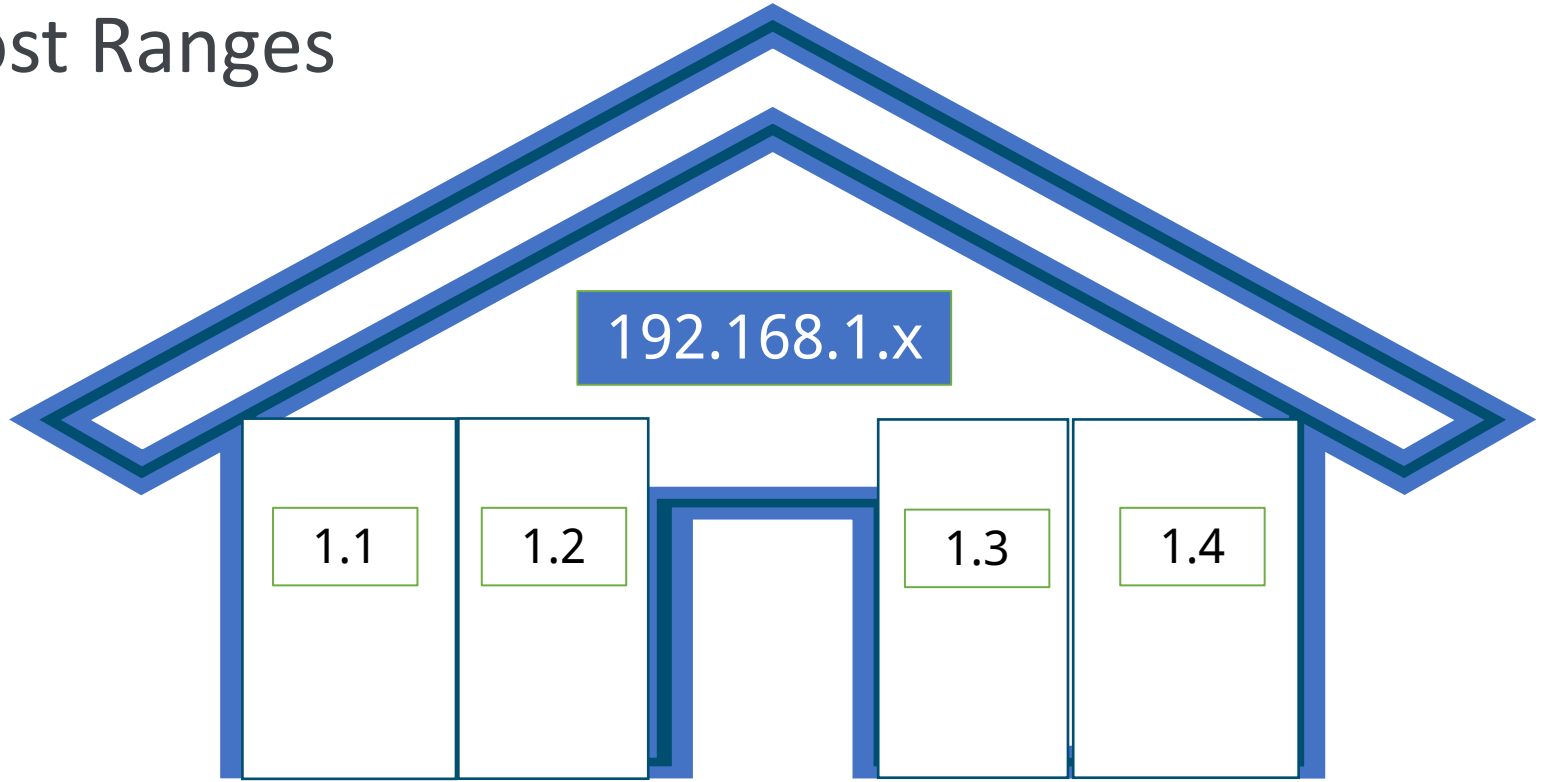
Host Ranges



Think of the “network” part of an IP address as the building’s address, and the “host” part as the apartment number.

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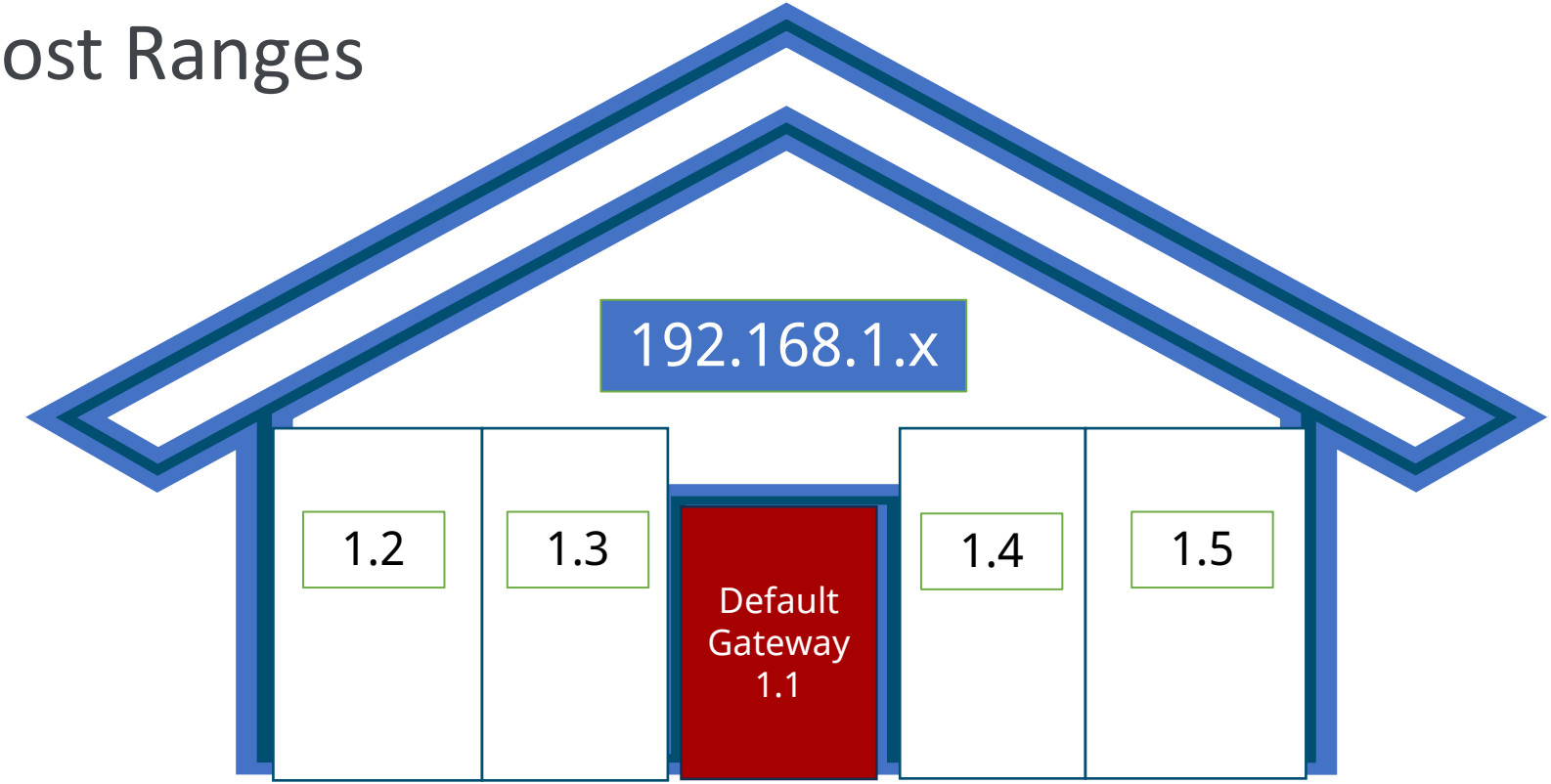
Host Ranges



The default gateway is like the main entrance of lobby of the apartment building

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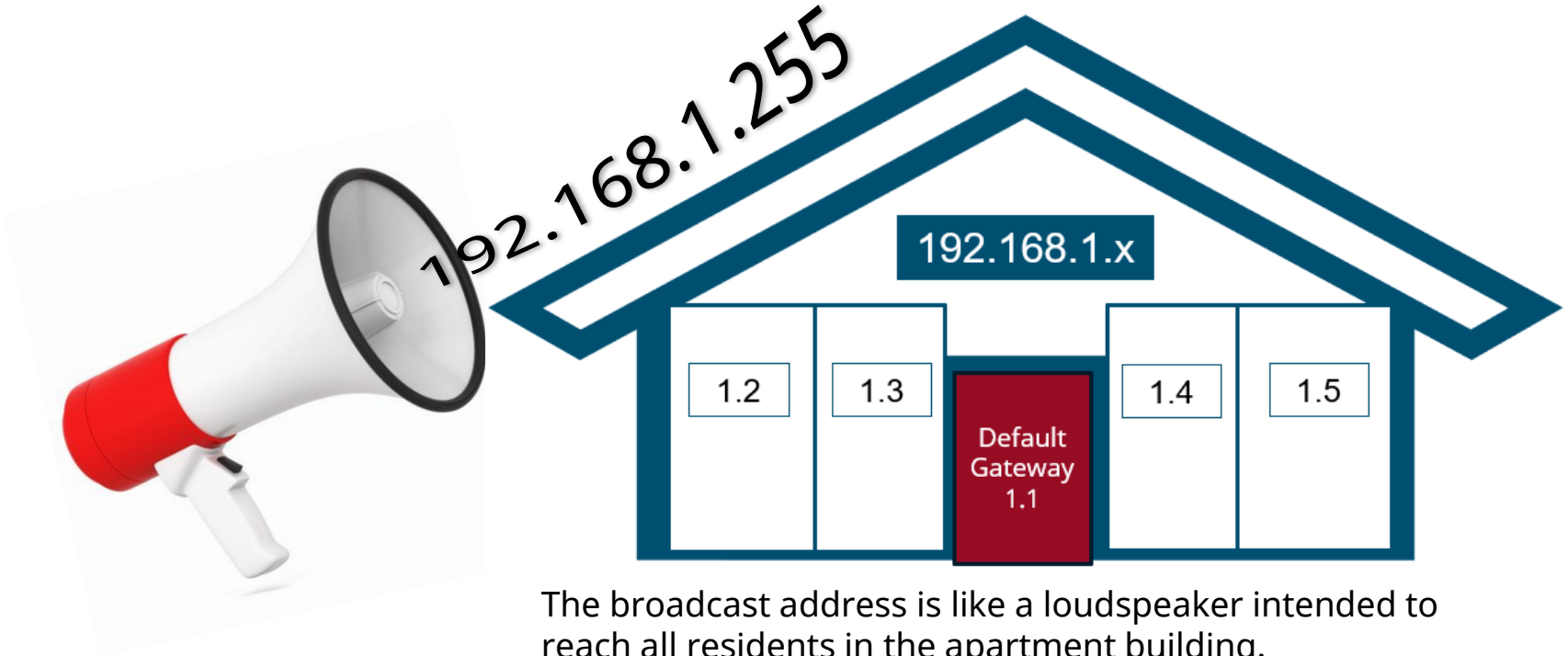
Broadcast Addresses



The broadcast address is like a loudspeaker intended to reach all residents in the apartment building.

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Activity: Fill in the Blanks

IP Address	Subnet Mask	Host Portion	Broadcast Address
192.168.2.123	255.255.255.0	.123	192.168.2.255
10.1.1.9	255.255.0.0	.1.9	10.1.255.255
198.51.100.16	255.255.255.0	.16	198.51.100.255

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IPv4 Subnetting

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Public vs. Private Addressing

Public IP Addressing	Private IP Addressing
• Assigned by ISPs • Unique identification on Internet • Devices requiring direct access to the Internet	• Assigned by private networks • Unique identification in private network • Devices not directly accessible from the Internet

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Private IP Addresses

Class	Range
A	10.0.0.0 to 10.255.255.255
B	172.16.0.0 to 172.31.255.255
C	192.168.0.0 to 192.168.255.255
D	224.0.0.0 to 239.255.255.255
E	240.0.0.0 to 255.255.255.255

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IPv4 Address Scheme Design

When planning a private IPv4 addressing scheme, consider:

- # of IP networks and subnetworks required
- # of hosts per subnet
- Network ID must be from a valid private range
- Network and host IDs cannot be all 1s (broadcast)
- Each host ID must be unique on subnet
- Each network ID must be unique on network

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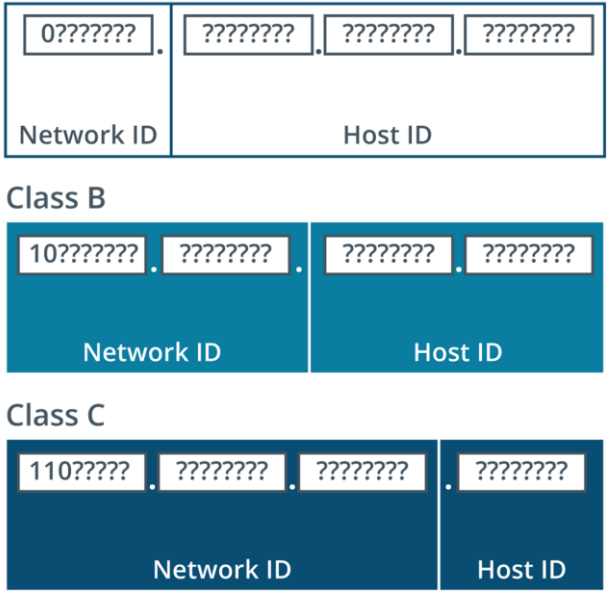
Classful Addressing



	Number of Networks	Number of Hosts per Network	First Octet of Address Range
Class A	126	16,777,214	1-126
Class B	16,384	65,534	128-191
Class C	2,097,152	254	192-223

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Classless Addressing

Borrowing Bits

- Would have represented host information to represent network information
  - Increase available network addresses
  - Reduce usable host addresses in each subnet

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Classless Addressing Example

Starting Network ID	192.168.89.0
Starting Subnet mask	255.255.255.0
Starting Network ID in binary	11000000.10101000.01011001.00000000

Borrow one bit from the host to give to the network	11000000.10101000.01011001.00000000
New network ID	192.168.89.128
Range of IP addresses	11000000.10101000.01011001.00000001 through 11000000.10101000.01011001.01111111 192.168.89.1 to 192.168.89.126
New subnet mask	255.255.255.128 or 192.168.89.128/25

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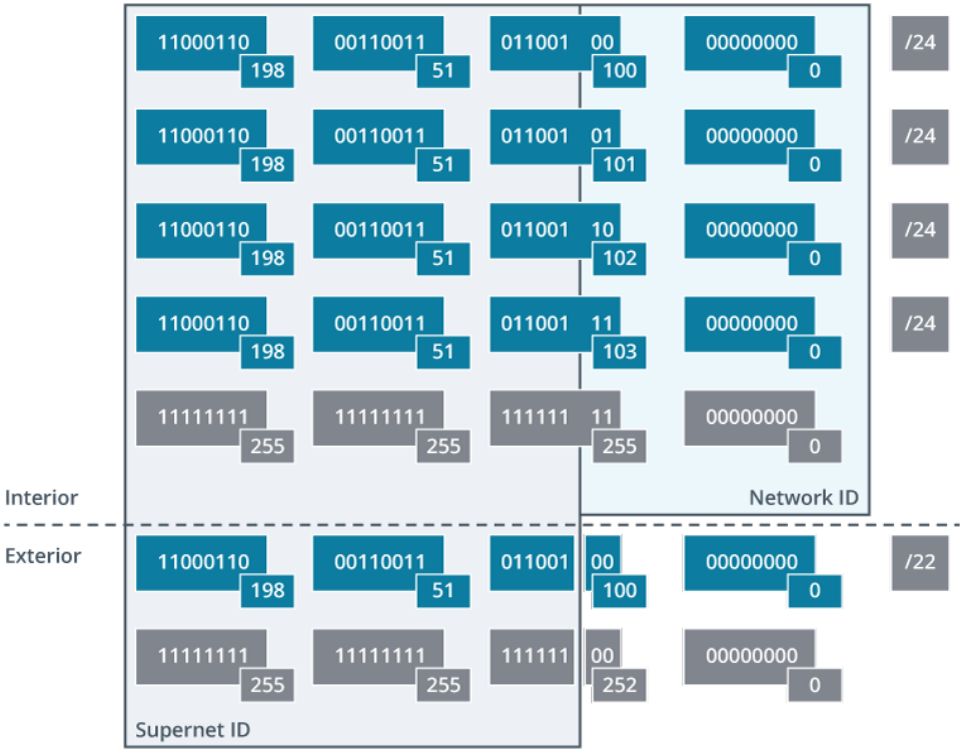
Classless Inter-Domain Routing (CIDR)

CIDR

- Allocates IP address space more efficiently
- Combines an IP address with a suffix (network portion of address)
- Allows for variable-length subnetting



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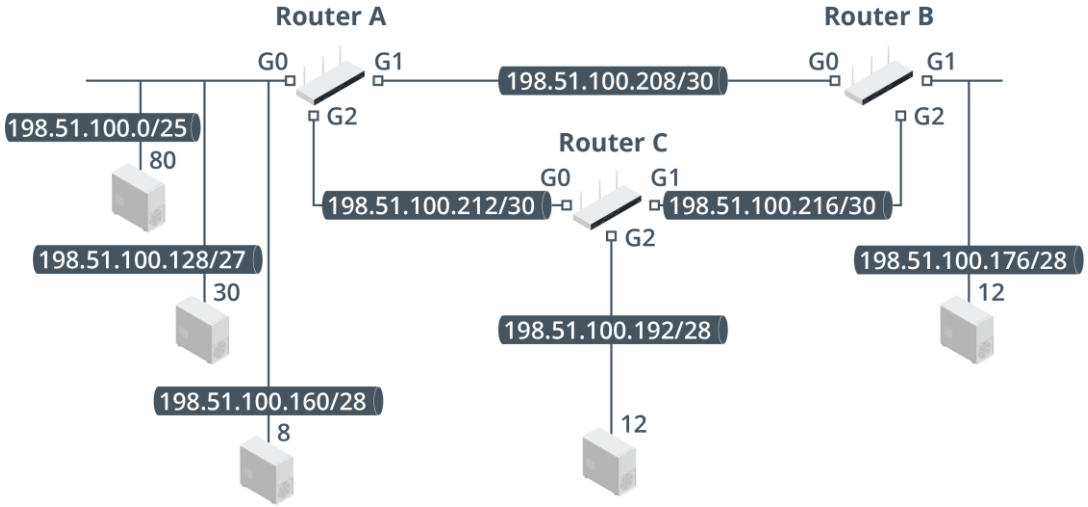
Variable Length Subnetting (VLSM)

VLSM

- Creates subnets of different sizes within the same network
- Enables efficient allocation of IP addresses
- Requires careful planning and management



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Activity: Classless Addressing

Starting Network ID	192.168.89.0
Starting Subnet mask	255.255.255.0
Starting Network ID in binary	11000000.10101000.01011001.00000000

Borrow two bits from the host to give to the network	11000000.10101000.01011001.00000000
New network ID	192.168.89.192
Range of IP addresses	192.168.89.1 to 192.168.89.62
New subnet mask	255.255.255.192 or 192.168.89.192 /26

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Lesson 4.4

IP Troubleshooting Tools

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ipconfig

ipconfig	ipconfig /all	ipconfig /renew	ipconfig /release
• View IP address, subnet mask, and default gateway for all network interfaces	• Also includes complete TCP/IP configuration parameters including DHCP and MAC address	• Renews the client's DHCPs leased IP address	• Releases the client's DHCPs leased IP address

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Ifconfig and ip

- Reports network interface configuration: **ip addr** (legacy is **ifconfig**)
- Single interface only: **ip addr show dev eth0**
- Status of interfaces: **ip link**
- Enable or disable an interface: **ip link set eth0 up|down**
- Modify the IP address configuration: **ip addr add|delete**

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arp

arp –a (or **arp –g**)

- Shows the ARP cache contents

arp –s *IPAddress MACAddress*

- Adds an entry to the ARP cache

arp –d

- Deletes all entries in the ARP cache

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ping

ping

- Tests connectivity with a given IP address
- Format: **ping** *IPaddress*

ping error messaging

- Destination host unreachable
- No reply (request timed out)

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Lesson 4.5

IP version 6

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IPv6 vs IPv4

Comparison	IP Version 4	IP Version 6
Address format	Dotted decimal format (192.168.1.1)	Hexadecimal format (2001:0db8:85a3:0000:0000:8a2e:0370:7334)
# of addresses	4.3 billion	340 undecillion (3.4 x 10^38)
Network and Host Portions	Changes based on subnet mask	First 64 bits are used for network ID. Second 64 bits are used for host ID

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IPv6 Address Format

Contains eight 16-bit numbers, with each double-byte expressed as 4 hex digits	
Binary Address	0010 0000 0000 0001 : 0000 1101 1011 1000 : 0000 0000 0000 0000 : 0000 0000 0000 0000 : 0000 1010 1011 1100 : 0000 0000 0000 0000 : 1101 1110 1111 0000 : 0001 0010 0011 0100
Hex Notation	2001:0db8:0000:0000:0abc:0000:def0:1234
Canonical Notation	2001:db8::abc:0:def0:1234

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IPv6 Unicast Addressing

Globally scoped unicast addresses are routable over the Internet and are the equivalent of public IPv4 addresses.

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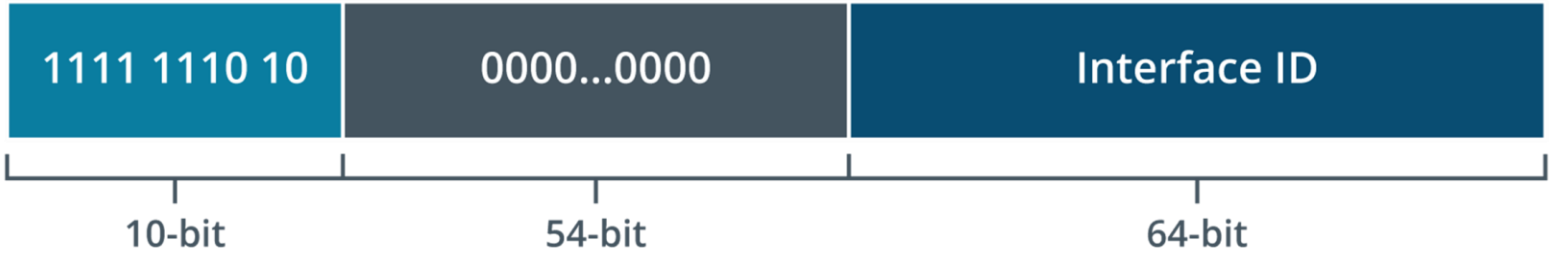
IPv6 Link Local Addressing

Link Local Addressing

- Span a single subnet and the equivalent of IPv4 private addressing
- Link local range is fe80::/10
- Start with a leading fe80, with the next 54 bits set to 0, and the last 64 bits are the interface ID



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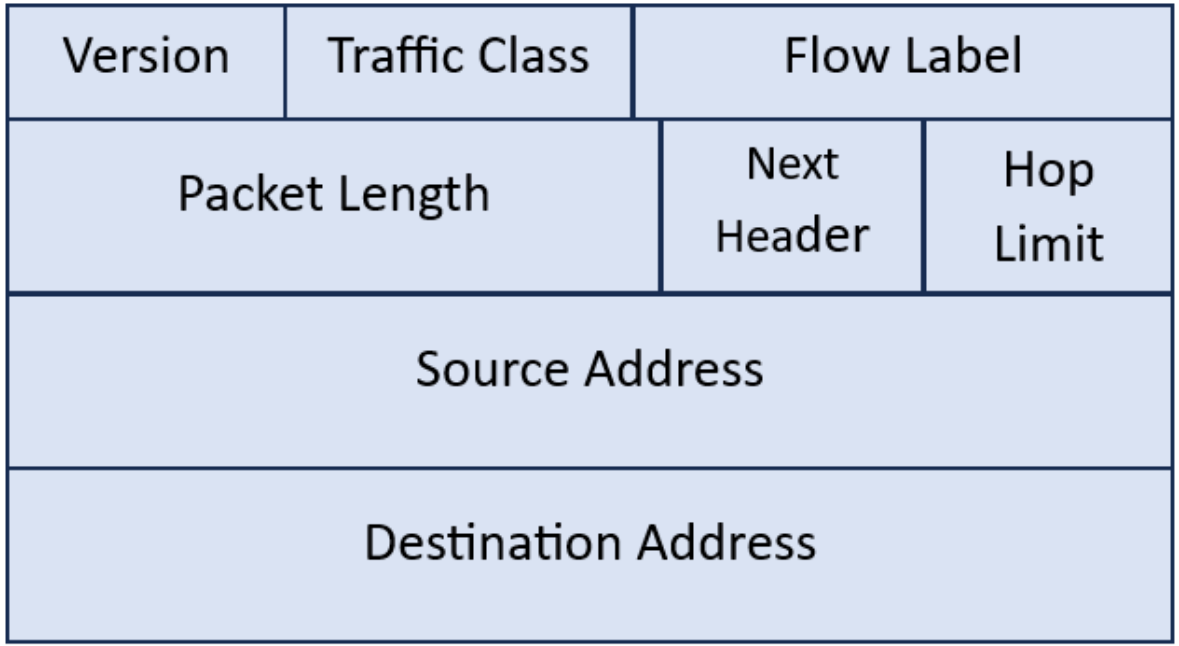
IPv6 Packets



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IPv6 Multicast and Anycast Addressing

Multicast

- First 8 bits: indicate the address is within the multicast scope
- Next 4 bits: used to flag types of multicast
- Next 4 bits: determine the scope
- Final 112 bits: define multicast groups within that scope

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IPv4 and IPv6 Transition Mechanisms

Dual Stack

- Allows IPv4 and IPv6 to run simultaneously
- Ideal for transitional periods where both IPv4 and IPv6 networks are operational

Tunneling

- Encapsulates IPv6 packets within IPv4 packets to traverse IPv4 networks
- Useful for IPv6 islands within an IPv4 network

NAT64

- Translates IPv6 into IPv4 and vice versa
- Best for environments transitioning to IPv6 but still needing access to IPv4 resources

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Common IPv6 Address Prefixes

Type	Prefix	Leading Hex Characters
Global unicast	2000::/3	2 3
Link local unicast	fe80::/10	fe80
Multicast	ff00::/8	ff
Multicast (link local)	ff02::/16	ff02::1 (all nodes) ff02::2 (all routers) ff02::1:2 (DHCP)
Solicited-node	ff02::1:ff00:0/104	ff02::1:ff
Unspecified	::/128	0::0
Loopback	::1/128	::1
Documentation/Examples	2001:db8::/32	2001:db8

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Summary

- **IP**: Creates interconnected networks (interwork) using unique addresses.
- **IPv4 Address**: 32-bit address divided into four octets.
- **Network Segment**: Represented by a subnet at layer 3 (Network.)
- **TCP/IP Utilities**: Command-line tools for network configuration and communication monitoring.
- **IPv6 Address**: Eight 16-bit numbers expressed in hexadecimal format.

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